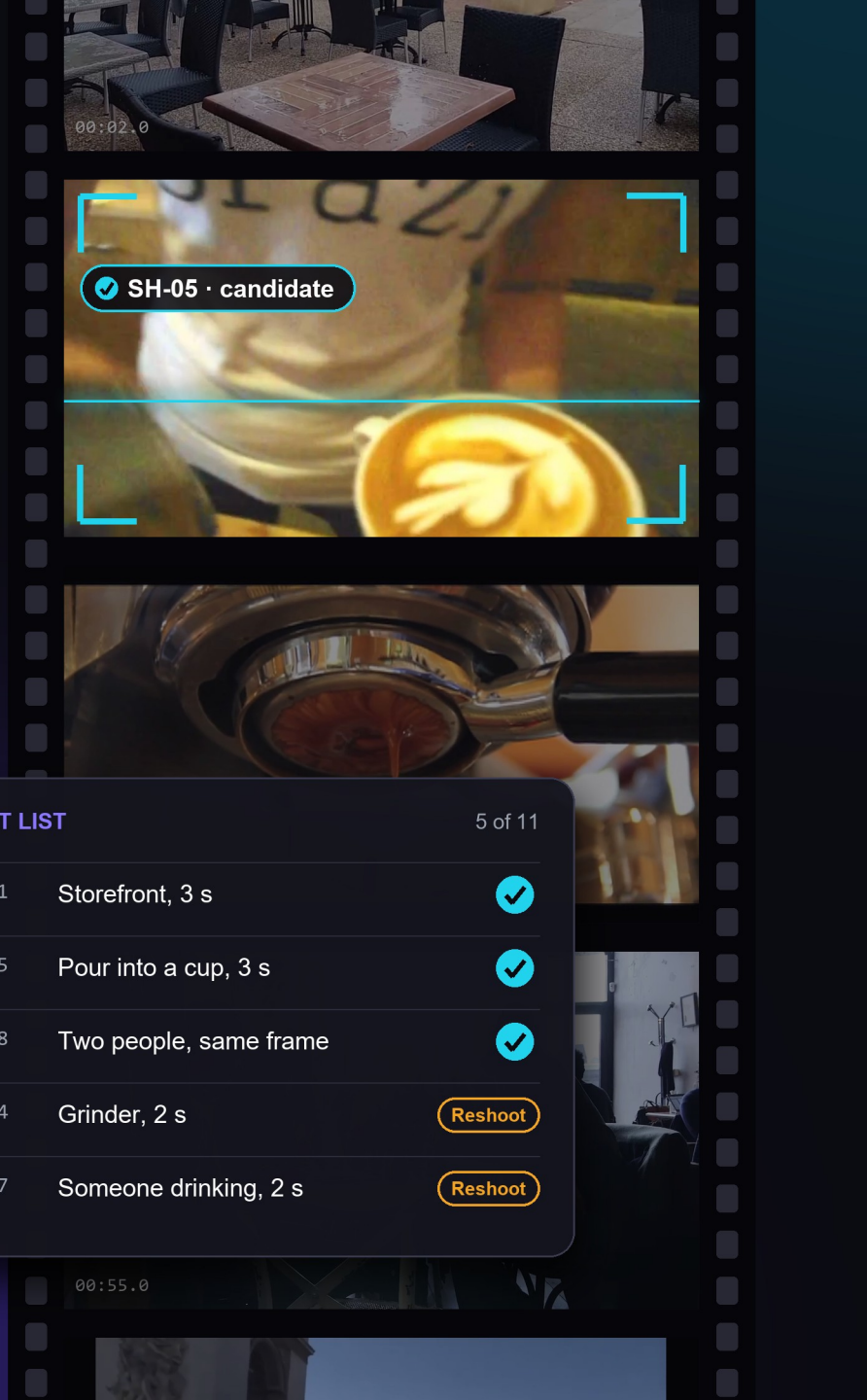
CREATOR AI · LIGHTNING TRACK · ASUS UGEN300

SceneGap Edge

Helps a crew check the shot list against the footage before leaving the location, with playable evidence and a person's decision on every shot.

- Runs on your own computer
- Evidence for every answer
- A person decides
- UGen300 port planned



Stage I proposal · [Team name] · github.com/[team]/scenegap-edge

PROBLEM

The missing shot is found at 23:00, after the café has closed



Illustrative scenario · two-person crew, café promo, 11 must-have shots

“We had pours. None of them was three seconds long. We found out in the edit.”

09:00	Shot list agreed with the client: storefront, interior, beans, grinder, 3 s pour, two barista takes, bag with readable brand.
17:40	Wrap. Several cards of footage; nobody has time to watch it all on site.
23:00	The editor scrubs for the pour: every take cuts away after 2 s, and nobody filmed a customer drinking.
Next week	Reshoot day: crew, café, client sign-off again.

PROBLEM

Existing tools find footage; SceneGap adds shot-list rules, evidence and reshoot tracking

TOOL	SEARCHES FOOTAGE	FINDS THE MOMENT IN A CLIP	SHOT-LIST RULES (LENGTH / TAKES / SAME FRAME)	RUNS LOCALLY
Premiere Pro media intelligence	yes	yes (In/Out range)	not in official docs	yes (on device)
DaVinci Resolve 21 IntelliSearch	yes	visual: whole clips	not in official docs	not stated
Final Cut Pro 12 visual search	yes	yes	not in official docs	not stated
Frame.io semantic search	yes (beta)	yes	not in official docs	no (cloud)
Jumper	yes	yes	not in official docs	yes
invideo AI script supervisor	matches footage to a shot list	not stated	done / pending	no (upload)
SceneGap Edge	**yes**	**yes**	**checked per rule; a person decides**	**yes**

> **The difference is the workflow around search: shot list → rules → evidence → a person's decision → reshoot list → re-check.**

Vendor documentation, accessed 2026-10-26 (Adobe media intelligence FAQ and others); URLs in docs/research/BUSINESS_FACTS.md.

PROBLEM

What works today, and what still needs proof

Shown today (live app, real footage)

- Shot list in the crew's words, with rules: length, takes, same frame
- Candidate clips with playable evidence on the original timecode
- Rule checks: 10 of 12 controlled cases judged right (slide 11)
- A person accepts, rejects, or adds the shot to the reshoot list
- New footage re-checked; only the new files are analysed

Not shown yet

- Reliable automatic missing-shot detection (0 of 38 on test, slide 14)
- Time saved for a real crew: no user study yet
- UGen300 speed and quality: not run on the device yet
- Human-checked labels: current labels are AI-assisted

SceneGap Edge today is a review assistant that keeps evidence and a person's decision; automation grows only as it is proven.

SOLUTION

From your shot list to a decision you can play back

01

Write the shot list

In Chinese, as the crew speaks. A person checks the English search phrase and adds rules: min. length, takes, same frame.

PC

02

Index footage locally

Find camera cuts and sample one frame per second on the original timecode; CLIP turns each frame into a searchable vector.

PC today → UGen300

03

Check each rule

Search finds candidate clips; each rule is checked and shown as pass, fail or can't tell. Failing clips stay visible.

PC

04

A person decides

Accept, reject with a reason, or add the shot to the reshoot list. Only the person's decision counts as done.

PC

UGen300 box = planned route; today the same model runs on the PC's CPU or GPU and is labelled cpu-/cuda-reference.

SOLUTION

The AI gives one of four answers; only a person can mark a shot done

The AI's answer and the person's decision are stored separately; only the person's counts as done.

Candidate

Every rule passes on at least one clip. Still needs a person to accept.

Partial

Something matched but a rule failed or can't be decided: too short, one take of two, a manual check left.

Not found

Nothing reached the threshold in the analysed files. Not proof of absence; a reshoot card is suggested.

Unverifiable

Files failed to decode or are missing, so absence cannot be claimed. Fix the scope first.

Person: accepted / rejected

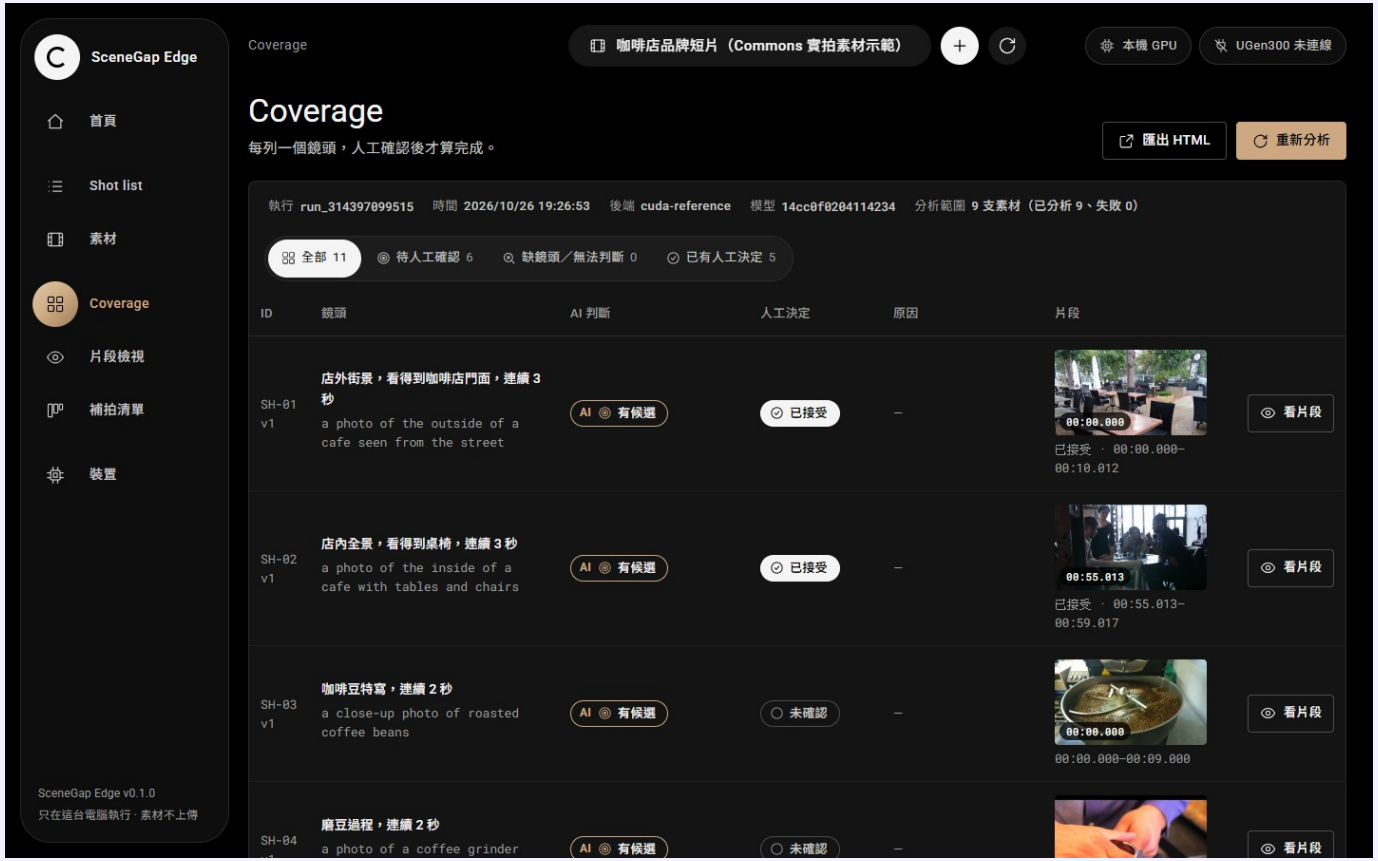
Bound to the shot's version. Edit the shot and the old decision no longer counts.

Outdated

Shot, footage, model or threshold changed since the run. The old answer is hidden, not reused.

SOLUTION

Before leaving, the crew sees which shots are confirmed and which need a look



Live app on 9 Wikimedia Commons clips after a CLIP run and a short review by a person. Frames are real footage.

Read in one pass

- Each row: the shot, the AI's answer, the person's decision, the clip
- Coverage counts only shots a person accepted
- Filters: waiting for review, missing, decided

SOLUTION

Every answer plays its clip, and a person decides



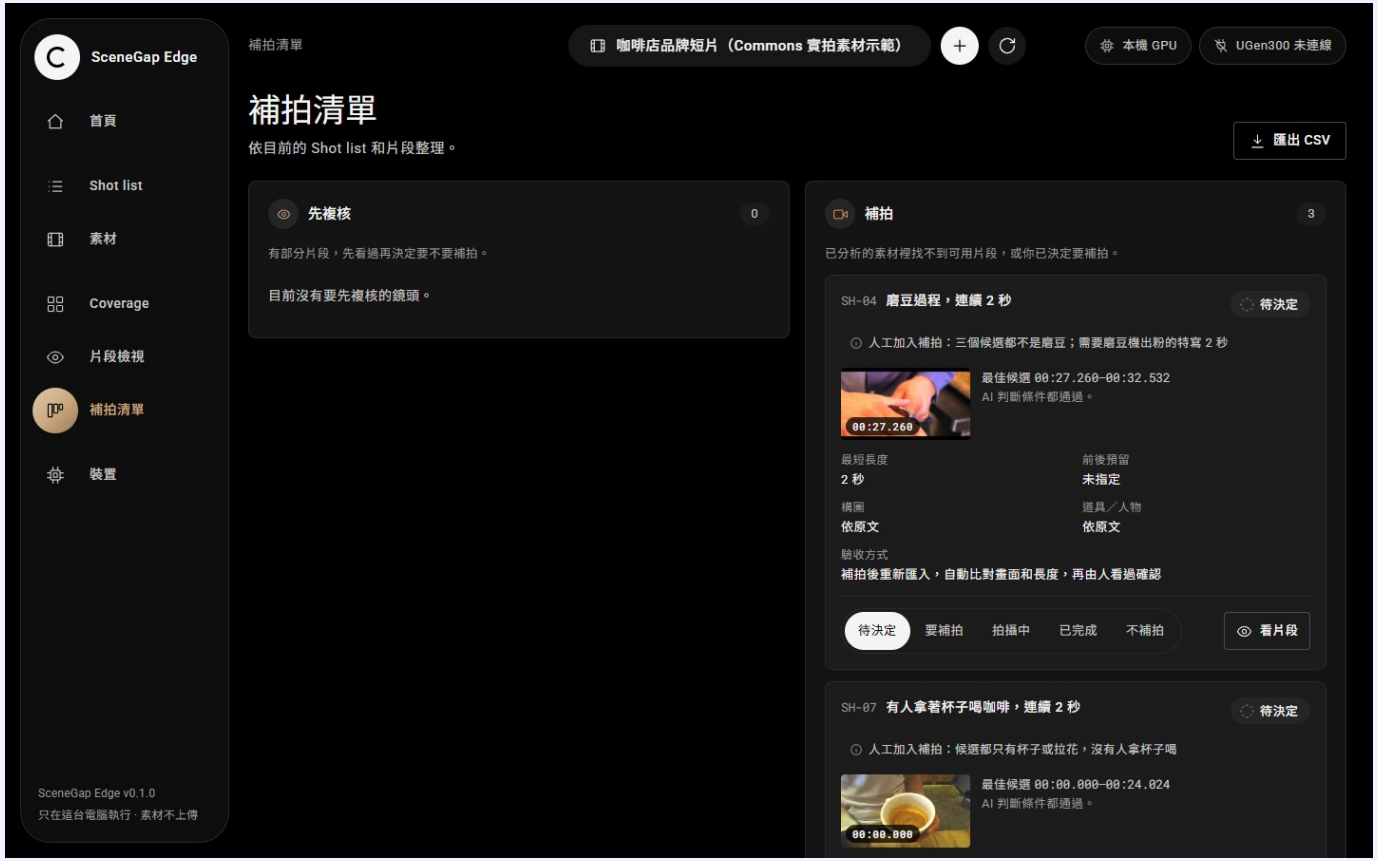
Shot SH-05 「咖啡倒入杯中或拉花，連續 3 秒」 (coffee or latte art poured into a cup, 3 s). The espresso stream with no cup was rejected; the latte-art pour accepted. The 24 s range is how long the frames stay close to the query, not a measured pour: the person checks the action.

One click per decision

- Accept, reject with a reason, adjust the range, add to reshoot list
- Filmstrip and clip ranges on the original timecode
- Every decision is logged and can be undone
- Original files are never touched

SOLUTION

Shots without a usable clip go on a reshoot list before the crew leaves



Same demo: the AI still offered candidates for the grinder and someone drinking; a person rejected them and put both, with the coffee bag, on the reshoot list.

What the crew gets

- The shot, why it is missing, the closest clip
- Minimum length and how it will be checked
- Status: to decide → reshoot → shooting → done

EVIDENCE

What each rule actually measures, and what it cannot tell

RULE	WHAT SCENEGAP MEASURES	WHAT IT CANNOT TELL	TODAY
Length	How long frames stay close to the query inside one camera cut (lower and upper bound; re-check every 0.25 s)	Whether the action itself lasts that long: a finished latte looks like a pour	auto, a person checks
Same frame	Both phrases above the threshold in the same sampled frame	Who is with whom; objects too small for CLIP	auto, a person checks
Takes	Distinct files, by content hash; duplicates of one file are one take	A re-encoded copy of the same take	auto, a person checks
Readable brand	Nothing automatic	Reading text	manual
Not found	Nothing reached the threshold in the analysed files	Proof that the shot was never filmed; failed files make it 'can't tell'	auto, scope shown

> **Dense sampling gives more observations, not certainty: a clip that looks like pouring is still checked by a person.**

EVIDENCE

Rule check on real footage: 10 of 12 cases right; misses are a finished latte and a re-encoded copy

CASE (REAL COMMONS FOOTAGE, CUT TO A KNOWN ANSWER)	SHOT RULE	SHOULD	SCENEGAP	RIGHT?
continuous 4 s latte-art pour	pour ≥ 3 s	pass	pass	yes
6 s pour from a different shoot	pour ≥ 3 s	pass	pass	yes
pour lasts only 2 s (rule: 3 s)	pour ≥ 3 s	not pass	not pass (duration_short)	yes
two 2 s pours split by another shot	pour ≥ 3 s	not pass	not pass (duration_short)	yes
finished latte, nothing is poured (5 s)	pour ≥ 3 s	not pass	pass	NO
two people at a café table, same frame	2 people in a café, same frame	pass	pass	yes
two people, but outdoors (not in a café)	2 people in a café, same frame	not pass	not pass (no_candidate)	yes
two people, then a café interior: separate shots	2 people in a café, same frame	not pass	not pass (no_candidate)	yes
one barista clip (rule: 2 takes)	barista, 2 takes	not pass	not pass (duration_short, takes_short)	yes
the same clip copied byte for byte	barista, 2 takes	not pass	not pass (duration_short, takes_short)	yes
the same clip re-encoded (still one take)	barista, 2 takes	not pass	pass	NO
two different barista clips	barista, 2 takes	pass	pass	yes

CLIP sees a finished latte as 'pouring'; takes are told apart by file hash. eval/rule_check.py, eval/results/rule_check.json; a separate experiment from the held-out test, frozen threshold.

EVIDENCE

How we tested: fixed protocol, open footage, AI-assisted labels

Step 1

Protocol first

Metrics, split and threshold rule committed to git before any label existed.

Step 2

Shot list

11 café-promo shots and queries written before any frame was seen.

Step 3

Footage

44 Wikimedia Commons clips, split by author: 18 dev, 26 test.

Step 4

AI-assisted labels

Two AI passes on 1 fps sheets agree (κ 0.91); a third pass on every disagreement. Human checks: Stage II.

Step 5

Test once

Threshold set on dev, frozen; test scored once. Versions, labels and result files are in the repo.

EVIDENCE

Finding footage: right clip in the top 5 for 7 of 11 shots; plain frame search found 8

METHOD	RIGHT CLIP IN TOP 5 (HIT@5)	TOP-5 PICKS THAT ARE RIGHT
Title keyword match (whole clips)	45% (5/11, 95% CI 21–72%)	36% (8/22, 95% CI 20–57%; fewer than 5 returned)
Single-frame semantic search	73% (8/11, 95% CI 43–90%)	40% (22/55, 95% CI 28–53%)
SceneGap: clips ranked after the rule checks	**64% (7/11, 95% CI 35–85%)**	**33% (18/55, 95% CI 22–46%)**

Lost SH-03 and SH-04, gained SH-06. The lost grinder clip passes every rule on its own (19 s run), so the rules did not reject it: ranking by a run's average similarity pushed it out. Stage II ranks by peak.

26 held-out clips, scored once, τ = 0.225. A hit needs 0.5 s overlap, so this layer does not test the length rule (slide 11 does).

EVIDENCE

Finding missing shots automatically does not work yet

Unit: one shot in one simulated 8-clip project (8 × 11 = 88). Missing = 38, present = 50. Found = 'candidate'; couldn't decide = 'partial' or 'unverifiable'.

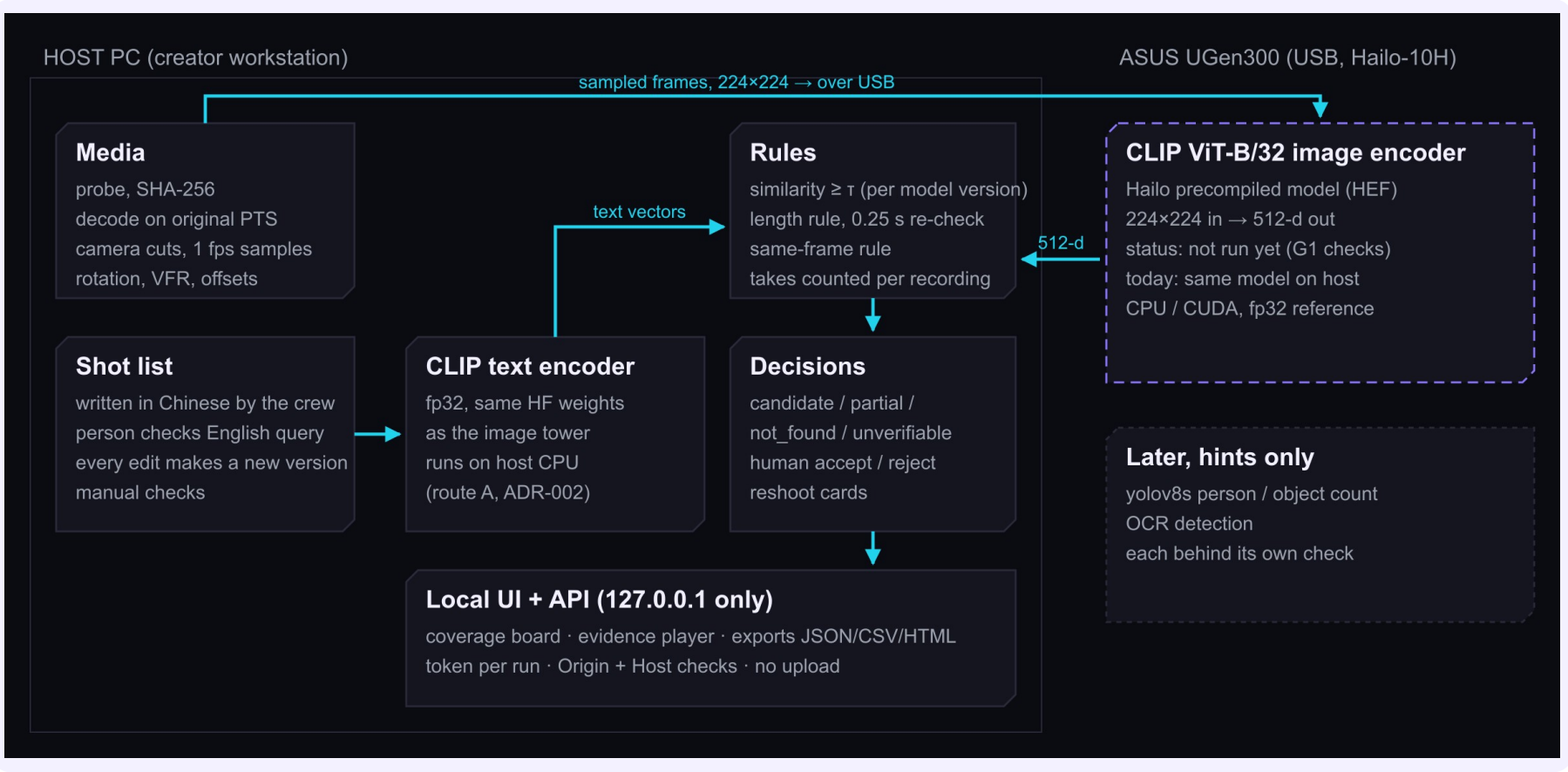
METHOD	MISSING SHOTS CAUGHT	RESHOOT ASKED, SHOT WAS THERE	COULDN'T DECIDE	SAID 'FOUND' WHEN ABSENT
Title keyword match	34/38 (89%)	22/50 (44%)	0/88 (0%)	4/38 (11%)
Single-frame semantic	0/38 (0%)	0/50 (0%)	0/88 (0%)	38/38 (100%)
SceneGap	**0/38 (0%)**	**0/50 (0%)**	**11/88 (12%)**	**30/38 (79%)**

> **SceneGap never said 'not found', so its 0% wrong reshoots means nothing yet. The threshold was tuned on single frames; Stage II calibrates per shot.**

eval/results/test_metrics.json; a dev-only sweep (relative score) catches 33% of gaps at 11% wrong reshoots, not used to rescore test.

HARDWARE

Plan: the image half of CLIP on the UGen300; the PC keeps the rest



Demo PC: Windows 11, RTX 4080. UGen300 supports Windows, Linux and Android (no macOS listed); on a Mac SceneGap runs on CPU only. Before the device replaces the PC model it must pass five G1 checks against the PC (appendix A1).

HARDWARE

On a desktop, analysis already costs 2.3–3.1 s per minute; UGen300's value is a hypothesis

BACKEND	DEVICE	ANALYSE (DECODE + MODEL)	SEARCH, TYPICAL / SLOWEST 5% (10K CLIPS)	MODEL LOAD
cpu-reference	desktop CPU	3.1 s per minute of video	11.77 / 14.29 ms	6.51 s
cuda-reference	NVIDIA GeForce RTX 4080	2.3 s per minute of video	7.61 / 13.39 ms	1.88 s
UGen300	Hailo-10H	not run yet	not run yet	not run yet

> **Stage II test: an editing app alone, then with CPU, GPU or UGen300 analysis — dropped frames, CPU/GPU load, power and index time.**

eval/results/perf.json (six short clips, ≤1280 px), interference.json. Playing a 1080p30 test pattern while analysing: 0 of 1,800 frames late (slowest 5%: 5.7 ms idle, 7.5 ms CPU, 6.2 ms GPU).

ENGINEERING

Engineering: 46 automated tests; 30 of 36 acceptance cases pass by test

These show the software is reliable, not that the AI is accurate. The other 6: two need a UGen300, one needs an object detector, three are measured, proxy or manual.

Media edge cases

Rotation, 29.97 fps, VFR, start offsets, corrupt and unsupported codecs, files still copying, moved files found again by hash.

Logic

'Can't tell' stays 'can't tell'; two short clips never count as one long one; duplicates are not two takes; editing a shot voids old decisions.

Security

127.0.0.1 only, per-run token, Origin + Host checks, authorised folders, CSV formula guard, escaped HTML, errors never expose internals.

Offline

Core flow tested with outbound sockets blocked. Footage never leaves the computer.

Failure

Disk full stops safely and resumes; interrupted jobs resume; originals' hashes unchanged.

Rebuild

A fresh venv from the public index passes all 46 tests; Dockerfile and CI included; Apache-2.0.

BUSINESS

Who pays: small crews and freelance editors who shoot on location

SIGNAL	NUMBER	SOURCE
US film and video editors / camera operators (2025)	39,400 / 32,600; 35% self-employed	US BLS OOH, accessed 2026-10-26
Median pay, editors (May 2025)	$75,420 / yr	US BLS OOH, accessed 2026-10-26
Day rate, small crew (estimate)	NT$12,000–28,000 (TW); $800–1,500 (US)	PRO360 quotes; rate cards
Creators worldwide	~50M, ~4% professional	Goldman Sachs (estimate, partly verified)
Shot-list practice	crews tick off shots and schedule pickup shots	StudioBinder guide

> **Value per project = hours saved × hourly cost + expected reshoot cost avoided − tool cost. Hours include setup, import, waiting and review; unmeasured until the user study.**

docs/research/BUSINESS_FACTS.md (URLs, access dates, verified flags).

BUSINESS

Open core; customers pay for a tested bundle and templates

Prices are hypotheses to test in interviews. Benchmarks: Resolve Studio $295 and Final Cut Pro $299.99 one-time; UGen300 reported at $299.99.

Free: Apache-2.0 core

Anyone can run, audit and extend the checker on CPU or GPU. Trust matters when footage never leaves the machine.

Paid: UGen300 bundle

Signed installer, tested driver and runtime versions, the device-checked model, and support. Sold with or for the accelerator.

Paid: shot-list templates

Per genre (café, product, event, wedding) with tested queries and rules; yearly updates.

Shown today, proven next

SceneGap Edge · a review assistant that keeps evidence and a person's decision for every shot

- Shown: shot list → candidate clips with evidence → a person's decision → reshoot list → re-check, on real footage
- Proof 1, the rules: more controlled cases, human-checked labels on the original video, ranking by peak
- Proof 2, people: 3–5 interviews, then crews checking a card by hand vs with SceneGap (time, misses, wrong reshoots)
- Proof 3, the UGen300: G1 checks on the device, then editing-app load with and without it

APPENDIX

A1 · G1: five checks before the UGen300 replaces the PC model

CHECK	MEASURE (DEV FRAMES, SAME PREPROCESSING)	PASS BAR	STATUS
G1a	Cosine, HEF vs fp32 embedding, per frame	median ≥ 0.98, p5 ≥ 0.95	not run yet
G1b	Spearman ρ of similarity per query	median ≥ 0.95, worst ≥ 0.85	not run yet
G1c	Top-5 / top-10 shot overlap	≥ 0.80 / ≥ 0.85	not run yet
G1d	Dev Hit@5, each backend with its own τ	no dev shot lost vs fp32	not run yet
G1f	Frames/s over USB, 60 s clip end to end	reported, no bar	not run yet
Harness	Same script, CPU vs CUDA fp32 (self-check)	≈ 1.0 expected	cos 1.0000, ρ 0.99999

> **Vendor: Hailo-10H ViT-B/32 image encoder, CIFAR-100 zero-shot 65.3 float → 64.5 on chip (PCIe, not USB).**

docs/MODEL_PLAN.md §2, §4; Hailo Model Zoo (vendor figures).

APPENDIX

A2 · What could break, and what we do then

RISK	FALLBACK
HEF fails G1 (quantisation hurts ranking)	Recalibrate with dev frames; TinyCLIP; else ship the CPU/GPU version and say so
A borderline camera cut splits a long take	The length check says 'can't tell', not 'too short'; the cut is shown
One global threshold fits some shots badly	One threshold per model version now; per type of shot next, set on dev
English phrase drifts from the Chinese shot	A person checks the phrase; a multilingual model only after its own check
AI-assisted labels, Commons footage, small test	Intervals shown; Stage II adds human labels and self-shot projects
Crews will not write shot lists	Templates; the interviews check this first

APPENDIX

A3 · Stage II order: rules and people first, then the device

Wk 1–2	Wk 2–3	Wk 3–5	Wk 4–6	Wk 6+
Rules and labels	**Missing shots**	**User study**	**UGen300**	**Chinese queries**
More controlled cases; a person labels sampled pairs on the original video; rank runs by peak.	Per-shot thresholds with a project-level target; preregister, score once on new footage.	3–5 interviews, then 8–12 crews: by hand vs SceneGap, counterbalanced; time, misses, wrong reshoots.	G1 on the device; editing-app load with CPU, GPU or UGen300 analysis.	A multilingual model (SigLIP 2), adopted only if it finds as many dev shots.

APPENDIX

A4 · Footage shown in this deck

#	WIKIMEDIA COMMONS FILE	AUTHOR	LICENCE
001	Vichy - Grand Café, terrasse sous la grêle.ogv	Romainbehar	CC0
007	Café Lomi, Paris - Vimeo.webm	ideafarm films	CC BY 3.0
012	Kaffe - Ystad-2024.ogg	Jonn Leffmann	CC BY 3.0
016	Lions Almelo zamelt koffie in voor Voedselbank.webm	AAVISIE	CC BY 3.0
019	MoliendoCafeEnLicuadora ohs02.webm	PetrohsW	CC BY-SA 4.0
020	Espresso video.ogv	Rosatrieu	CC BY-SA 3.0
026	Latte art flower - 01.ogv	Anna Frodesiak	CC0
032	Sugar falling into coffee.webm	Anselm Schüler	CC BY-SA 4.0
041	Empty Streets of Paris COVID19.webm	CreatorsCloud	CC BY 3.0

Screenshots and cover frames come from these 9 clips. All 44 clips used in the evaluation, with links: docs/CREDITS.md in the repository.